CUTTING OUT THE COHEN-MACAULAY LOCUS FOR A GROUPOID WITH FLAT FINITELY PRESENTED STRUCTURE MAPS, YIELDING A COHEN-MACAULAY RESTRICTION WITH SURJECTIVE STRUCTURE MAPS

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Allowed standard prerequisites (editorial).

- The local-source descent lemma for the largest P-open along a base change with section, in the P/Q/R framework (more-groupoids chapter, Lemma lemma-local-source, canonical tag 04LH; collected on this lane as L22513). cited at proof lines 863-871. Applied with g = id, Q = locally of finite presentation, R = flat and locally of finite presentation, P = Cohen-Macaulay to produce U' with the maximality property (1).
- The local-source-apply remark instantiating the framework for morphism groupoids (more-groupoids chapter, Remark remark-local-source-apply, canonical tag 047W). cited at proof lines 869-870. Justifies the application of the local-source lemma in this setting.
- The Cohen-Macaulay locus open of a flat finitely presented morphism is dense in every fibre (more-morphisms chapter, Lemma more-morphisms-lemma-flat-finite-presentation-CM-open, canonical tag 045U). cited at proof lines 879-883. Provides fibre density of the CM-locus opens, which with the identity section gives the surjectivity conclusions (3) and (4).
- The restriction of a groupoid to an invariant open carries a groupoid structure (groupoids chapter, Section groupoids-section-restrict-groupoid, canonical tag 02VA; kind reference). cited at proof lines 884-886. The chapter discussion of restrictions underlies part (5): R' = s^{-1}(U') intersect t^{-1}(U') defines the restricted groupoid (U', R', s', t', c').

Source reference. Original lemma 04LH: sources/stacks-chow/tags.txt, source lines 6008–6008. Cited at proof lines 863-871: the applied framework producing U' with the largest s-CM-open property (1).

Source reference. Original remark 047W: sources/stacks-chow/tags.txt, source lines 5523–5523. Cited at proof lines 869-870: instantiates the local-source application in this setting.

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Source reference. Original lemma 045U: sources/stacks-chow/tags.txt, source lines 5449–5449. Cited at proof lines 879-883: fibre density of the CM-locus open, giving surjectivity (3) and (4) with the identity section.

Source reference. Original reference 02VA: sources/stacks-chow/tags.txt, source lines 3761–3761. Cited at proof lines 884-886: the restriction discussion underlying part (5), the groupoid structure on R'.

Lemma 0.2. *Let S be a scheme. Let (U, R, s, t, c) be a groupoid over S. Assume s and t are flat and locally of finite presentation. Then there exists an open $U' \subset U$ such that*

1. *$t^{-1}(U') \subset R$ is the largest open subscheme of R on which the morphism s is Cohen-Macaulay,*
2. *$s^{-1}(U') \subset R$ is the largest open subscheme of R on which the morphism t is Cohen-Macaulay,*
3. *the morphism $t|_{s^{-1}(U')} : s^{-1}(U') \to U$ is surjective,*
4. *the morphism $s|_{t^{-1}(U')} : t^{-1}(U') \to U$ is surjective, and*
5. *the restriction $R' = s^{-1}(U') \cap t^{-1}(U')$ of R to U' defines a groupoid (U', R', s', t', c') which has the property that the morphisms s' and t' are Cohen-Macaulay and locally of finite presentation.*

Proof. Apply Lemma 04LH with $g = \mathrm{id}$ and $\mathcal{Q}$ ="locally of finite presentation", $\mathcal{R}$ ="flat and locally of finite presentation", and $\mathcal{P}$ ="Cohen-Macaulay", see Remark 047W. This gives us an open $U' \subset U$ such that Let $t^{-1}(U') \subset R$ is the largest open subscheme of R on which the morphism s is Cohen-Macaulay. This proves (1). Let $i : R \to R$ be the inverse of the groupoid. Since i is an isomorphism, and $s \circ i = t$ and $t \circ i = s$ we see that $s^{-1}(U')$ is also the largest open of R on which t is Cohen-Macaulay. This proves (2). By More on Morphisms, Lemma 045U the open subset $t^{-1}(U')$ is dense in every fibre of $s : R \to U$. This proves (3). Same argument for (4). Part (5) is a formal consequence of (1) and (2) and the discussion of restrictions in Groupoids, Section 02VA. □